

Historic Kinematic Solar Cycle Summary

Generated on: 2026-06-12

Cycle	Start	Max	Amplitude	Kinematic Event	Phase
1	1755-06	1761-06	137.6	1762-08	Decreasing Phase
2	1766-06	1769-11	188.8	1771-09	Decreasing Phase
3	1775-06	1778-06	259.9	1783-07	Decreasing Phase
4	1784-07	1788-01	231.7	1792-08	Decreasing Phase
5	1798-03	1804-11	81.2	1802-01	Rising Phase
6	1810-05	1816-04	79.8	1812-05	Rising Phase
7	1823-02	1830-02	115.8	1831-06	Decreasing Phase
8	1833-11	1837-03	239.7	1843-08	Decreasing Phase
9	1843-11	1848-06	215.7	1851-08	Decreasing Phase
10	1856-02	1860-02	183.4	1859-04	Rising Phase
11	1867-04	1870-09	233.4	1873-01	Decreasing Phase
12	1878-11	1883-12	120.6	1881-08	Rising Phase
13	1890-02	1893-11	144.1	1891-03	Rising Phase
14	1901-09	1905-11	102.6	1902-09	Rising Phase
15	1913-07	1917-09	171.4	1920-01	Decreasing Phase
16	1923-05	1928-05	125.8	1932-06	Decreasing Phase
17	1933-10	1937-06	191.8	1941-01	Decreasing Phase
18	1944-03	1947-06	213.6	1950-12	Decreasing Phase
19	1954-05	1957-12	281.6	1961-10	Decreasing Phase
20	1964-10	1968-11	154.0	1970-10	Decreasing Phase

Cycle	Start	Max	Amplitude	Kinematic Event	Phase
21	1976-07	1979-12	229.4	1982-05	Decreasing Phase
22	1986-07	1989-08	212.5	1991-04	Decreasing Phase
23	1996-07	2002-01	179.3	1999-10	Rising Phase
24	2008-12	2014-04	113.6	2011-04	Rising Phase

Statistical Test: Rising vs. Decreasing Phase

Metric	Rising Phase	Decreasing Phase
Count	8	16
Mean Amplitude	125.6	200.2
Median Amplitude	117.1	213.1

The Amplitude Ceiling (Median Analysis)

Because standard means can be skewed by historical outliers, comparing the median amplitude provides a highly robust metric. Cycles experiencing a natural kinematic reversal achieve a median amplitude of **213.1**, whereas cycles prematurely disrupted in their Rising Phase are constructively capped at a median of **117.1**. This represents a **45.0% systemic suppression** of the solar dynamo.

Parametric Validation (Welch's T-Test)

- **T-Statistic:** -3.9971
- **P-Value:** 0.00048

Advanced Validation (Non-Parametric Permutation Test)

To rigorously validate this delta without relying on Gaussian assumptions, we conducted a permutation test.

- **Resamples:** 10,000
- **Observed Difference in Means:** 74.59 sunspots
- **Empirical P-Value:** 0.00070

Conclusion: The result is statistically significant. It is mathematically improbable ($p < 0.05$) that the observed 45.0% amplitude suppression occurred by random chance.

Sensitivity Analysis: Leave-One-Out (Jackknife)

To ensure the statistical significance and amplitude suppression are not artificially driven by a single historical outlier, we performed a Leave-One-Out (Jackknife) resampling analysis.

- **Worst-Case P-Value:** *0.00220 (Occurs if Cycle 5 is removed)*
- **Worst-Case Suppression:** *43.4% (Occurs if Cycle 5 is removed)*

Conclusion: The model is **HIGHLY ROBUST**. Even in the absolute worst-case scenario (dropping the most influential historical cycles), the mathematical probability remains significant and the physical amplitude ceiling holds.